Homework 00.11.26 – Exercises 31–37

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Exercise 31 (Decomposition of Bregman divergence)

Let $f = \frac{1}{n}\sum_{i=1}^n f_i$. Prove that

$$D_f(x,y) = \frac{1}{n}\sum_{i=1}^n D_{f_i}(x,y).$$

Proof. We start from the definition of the Bregman divergence of f and substitute $f = \frac{1}{n}\sum_{i=1}^n f_i$. Since the gradient is linear, we also have $\nabla f(y) = \frac{1}{n}\sum_{i=1}^n \nabla f_i(y)$, so

$$D_f(x,y) = f(x) - f(y) - \langle \nabla f(y), x-y\rangle = \frac{1}{n}\sum_{i=1}^n f_i(x) - \frac{1}{n}\sum_{i=1}^n f_i(y) - \left\langle \frac{1}{n}\sum_{i=1}^n \nabla f_i(y), x-y \right\rangle.$$

The inner product is linear in its first argument, so the last term is also an average over i. Collecting the terms with the same index i under one sum, we see that each bracket is exactly the Bregman divergence of f_i:

$$D_f(x,y) = \frac{1}{n}\sum_{i=1}^n \underbrace{\big(f_i(x) - f_i(y) - \langle \nabla f_i(y), x-y\rangle\big)}_{D_{f_i}(x,y)} = \frac{1}{n}\sum_{i=1}^n D_{f_i}(x,y). \qquad \square$$

Exercise 32 (Theorem 38)

Assume that

1. f is μ-convex.
2. Each f_i is convex and L_i-smooth.

Then SGD-NS with stepsize $0 < \gamma \le \frac{1}{2A''}$, where $A'' = \max_i \frac{L_i}{np_i}$, satisfies

$$\mathbb{E}\big[\left\|x^k - x^\star\right\|^2\big] \le (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{2\gamma\sigma_\star^2}{\mu},$$

where

$$\sigma_\star^2 = \mathrm{Var}\big[g(x^\star)\big] = \mathbb{E}\big[\|g(x^\star)\|^2\big] - \left\|\mathbb{E}\big[g(x^\star)\big]\right\|^2 = \left(\frac{1}{n^2}\sum_{i=1}^n \frac{\|\nabla f_i(x^\star)\|^2}{p_i}\right) - \|\nabla f(x^\star)\|^2.$$

Proof. The theorem is a consequence of Theorem 33 and Lemma 37, so the plan is to check that the assumptions of Theorem 33 hold for SGD-NS and to compute the constants A and C. The function f is μ-convex by assumption, and the stochastic gradient $g(x) = \frac{\nabla f_i(x)}{np_i}$ of SGD-NS is unbiased, since $\mathbb{E}\big[g(x)\big] = \sum_{i=1}^n p_i \frac{\nabla f_i(x)}{np_i} = \nabla f(x)$. So it remains to check the AC assumption.

From Lemma 37, using the convexity and L_i-smoothness of f_i, we have that

$$\mathbb{E}\left[\|g(x) - g(y)\|^2\right] \leq 2A'' D_f(x,y), \qquad \text{where } A'' = \max_i \frac{L_i}{np_i}.$$

Moreover, for $G(x,y) = \mathbb{E}\left[\|g(x) - \nabla f(y)\|^2\right]$ we have

$$G(x,y) \leq 4A'' D_f(x,y) + 2\operatorname{Var}\left[g(y)\right] \qquad \forall x, y \in \mathbb{R}^d,$$

which for $x = x^k$ and $y = x^\star$ is the AC assumption.

So, substituting everything into Theorem 33, we get

$$\mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \leq (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{\gamma C}{\mu},$$

where $0 < \gamma \leq \frac{1}{A} = \frac{1}{2A''}$ with

$$A = 2A'' = 2\max_i \frac{L_i}{np_i} \qquad \text{and} \qquad C = 2\operatorname{Var}\left[g(x^\star)\right] = 2\sigma_\star^2. \qquad \square$$

Exercise 33

Prove that

$$\sigma_\star^2 = \mathbb{E}\left[\|g(x^\star) - \nabla f(x^\star)\|^2\right] = 0 \quad \text{iff} \quad \nabla f_i(x^\star) = np_i \nabla f(x^\star) \quad \forall i = 1, \ldots, n.$$

Proof. We use the formula for $\sigma_\star^2$ from Exercise 32,

$$\sigma_\star^2 = \left(\frac{1}{n^2}\sum_{i=1}^n \frac{\|\nabla f_i(x^\star)\|^2}{p_i}\right) - \|\nabla f(x^\star)\|^2,$$

and prove the two directions separately.

1) (⇐) Assume that $\nabla f_i(x^\star) = np_i \nabla f(x^\star)$ for all i. Substituting this into the formula, we get $\|np_i \nabla f(x^\star)\|^2 / p_i = n^2 p_i \|\nabla f(x^\star)\|^2$, so the factor $\frac{1}{n^2}$ cancels, and since the p_i sum up to one, the two terms are equal:

$$\begin{aligned}\left(\frac{1}{n^2}\sum_{i=1}^n \frac{\|\nabla f_i(x^\star)\|^2}{p_i}\right) - \|\nabla f(x^\star)\|^2 &= \left(\frac{1}{n^2}\sum_{i=1}^n \frac{\|np_i \nabla f(x^\star)\|^2}{p_i}\right) - \|\nabla f(x^\star)\|^2 \\ &= \left(\underbrace{\sum_{i=1}^n p_i}_{1}\right)\|\nabla f(x^\star)\|^2 - \|\nabla f(x^\star)\|^2 = 0 \;\Rightarrow\; \sigma_\star^2 = 0.\end{aligned}$$

2) (⇒) Let $\sigma_\star^2 = 0$, and let's assume that there are $p_i' > 0$ such that $\sum_i p_i' = 1$ and $\nabla f_i(x^\star) = np_i' \nabla f(x^\star)$. If $\nabla f(x^\star) = 0$, then $\nabla f_i(x^\star) = 0 = np_i \nabla f(x^\star)$ and there is nothing to prove, so let $\nabla f(x^\star) \neq 0$. Substituting $\nabla f_i(x^\star)$ into the formula and taking $\|\nabla f(x^\star)\|^2$ out of the brackets, we get

$$\sigma_\star^2 = \|\nabla f(x^\star)\|^2 \left(\frac{1}{n^2}\sum_{i=1}^n \frac{(p_i' n)^2}{p_i} - 1\right) = 0 \iff \sum_{i=1}^n \frac{p_i'^2}{p_i} = 1.$$

One such solution for p_i' is $p_i' = p_i$. It also turns out that it is the only solution: indeed, using Exercise 38 with $a_i := p_i'^2$ and $p_i := p_i$, we get the result. $\square$

Exercise 34

Prove part (iii) of Lemma 44.

Lemma 44. Let $f_1, \dots, f_n : \mathbb{R}^d \to \mathbb{R}$ be differentiable and $f(x) := \frac{1}{n}\sum_{i=1}^n f_i(x)$.

(i) If f_i are L_i-smooth, then f is L-smooth with $L \le \frac{1}{n}\sum_{i=1}^n L_i$.

(ii) If f is L-smooth and f_i are convex, then f_i is L_i-smooth with $L_i \le nL$.

(iii) If f is L'-smooth and f_i are convex and L_i'-smooth, then there exist $L \le L'$ and $L_i \le L_i'$ $\forall i$ such that f is L-smooth, f_i is L_i-smooth $\forall i$, and $L \le \max_i L_i \le nL$.

Proof of (iii). The idea is to take the smallest smoothness constants of f and f_i and then to apply parts (i) and (ii) to them. Let's take

$$L = \min\Big(L',\ \inf\big\{L^* \in \mathbb{R}^+ : f \text{ is } L^*\text{-smooth}\big\}\Big)$$

and

$$L_i = \min\Big(L_i',\ \inf\big\{L_i^* \in \mathbb{R}^+ : f_i \text{ is } L_i^*\text{-smooth}\big\}\Big).$$

Now, using (i) for the L_i-smooth functions f_i, we get that f is L''-smooth and $L'' \le \frac{1}{n}\sum_i L_i$, but because $L \le L''$ (recall that L is the smallest smoothness constant of f), we get that

$$L \le \frac{1}{n}\sum_{i=1}^n L_i \le \max_i L_i.$$

Now, using (ii) for the L-smooth function f and the convex functions f_i, we get that f_i is L_i''-smooth and $L_i'' \le nL$, but because $L_i \le L_i''$ (again, L_i is the smallest smoothness constant of f_i), we get that $L_i \le nL$, so $\max_i L_i \le nL$, which proves Lemma 44 (iii):

$$L \le \frac{1}{n}\sum_{i=1}^n L_i \le \max_i L_i \qquad \text{and} \qquad \max_i L_i \le nL. \qquad \square$$

Exercise 35

Give an example of L_i-smooth functions f_i for which $\frac{1}{n}\sum_i L_i \ll \max_i L_i$.

Solution. The idea is to make one function "much less smooth" than all the others: one large L_1 is enough to make $\max_i L_i$ large, while in the average it is divided by n. Let $n = 1000$ and

$$f_1 = 500x^2, \qquad f_2 = x^2, \qquad f_i = \tfrac{1}{2}x^2 \quad \forall i = 3, \dots, n.$$

A function cx^2 with $c > 0$ has the derivative $2cx$, which is $2c$-Lipschitz, so it is $2c$-smooth. Hence $L_1 = 1000$, $L_2 = 2$ and $L_i = 1$ for $i \ge 3$. In this case

$$\frac{1}{n}\sum_{i=1}^n L_i = \frac{1}{1000}\cdot\Big(1000 + 2 + \sum_{i=3}^{1000} 1\Big) = \frac{2000}{1000} = 2$$

and $\max_i L_i = 1000$, so we have

$$\max_i L_i = 1000 \gg 2 = \frac{1}{n}\sum_{i=1}^n L_i. \qquad \square$$

Exercise 36

Let $a_1, \dots, a_n > 0$. Prove that the solution to the optimization problem

$$D_{\Delta_n} := \min_{p \in \Delta_n} \max_i \frac{a_i}{p_i}$$

is given by $p_i = \frac{a_i}{\sum_j a_j}$ for all $i = 1, \dots, n$.

Proof. The proof consists of two steps: first we compute the value of the objective at p^* (here it is a partition that is defined in the exercise), and then we show that any other $p' \in \Delta_n$ gives a strictly larger value. Note that $p^* \in \Delta_n$, since all $p_i^* > 0$ and $\sum_i p_i^* = 1$.

First of all, note that for $p_i^* = \frac{a_i}{\sum_j a_j}$ we have

$$D := \max_i \frac{a_i}{p_i^*} = \max_i \Big(\sum_j a_j \Big) = \sum_j a_j,$$

i.e. all the ratios $\frac{a_i}{p_i^*}$ are equal to the same number $\sum_j a_j$.

Let's take another $p' \in \Delta_n$ and calculate $D' = \max_i \frac{a_i}{p_i'}$. If $p' \neq p^*$, then there is an i with $p_i' \neq p_i^*$. If $p_i' < p_i^*$, then let's assign $k = i$, and if $p_i' > p_i^*$, then, since $\sum_j p_j' = \sum_j p_j^* = 1$, there is $j \neq i$ with $p_j' < p_j^*$, and we will assign $k = j$. So in both cases there is a k with $p_k' < p_k^*$. Since $a_k > 0$ and $0 < p_k' < p_k^*$, decreasing the denominator increases the ratio, and we get

$$D = \sum_j a_j = \frac{a_k}{p_k^*} < \frac{a_k}{p_k'} \leq \max_i \frac{a_i}{p_i'} = D'.$$

So for any $p' \neq p^*$ we have $D < D'$, which means that p^* is the (unique) solution. □

Exercise 37 (Importance sampling 2 – simple version)

Let $a, b > 0$. Prove that the solution to the optimization problem

$$\min_{0<p<1} \frac{a}{p} + \frac{b}{1-p}$$

is given by $p = \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}$.

Proof. We find the minimiser by looking at the first and second derivatives of the objective. Let $\varphi(p) := \frac{a}{p} + \frac{b}{1-p}$ for $p \in (0,1)$. Then

$$\varphi'(p) = -\frac{a}{p^2} + \frac{b}{(1-p)^2}, \qquad \varphi''(p) = \frac{2a}{p^3} + \frac{2b}{(1-p)^3} > 0,$$

so φ is strictly convex on $(0,1)$, and a point where φ' equals 0 is its unique minimiser. Equating the derivative to 0 and using $p > 0$, $1 - p > 0$ to take square roots, we get

$$\varphi'(p) = 0 \iff \frac{a}{p^2} = \frac{b}{(1-p)^2} \iff \frac{\sqrt{a}}{p} = \frac{\sqrt{b}}{1-p} \iff \sqrt{a}\,(1-p) = \sqrt{b}\,p \iff p = \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}.$$

Since $a, b > 0$, this p lies in $(0,1)$, so it is the solution. Using $1 - p = \frac{\sqrt{b}}{\sqrt{a}+\sqrt{b}}$, the optimal value is

$$\varphi\left(\frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}\right) = \sqrt{a}\,\big(\sqrt{a}+\sqrt{b}\big) + \sqrt{b}\,\big(\sqrt{a}+\sqrt{b}\big) = \big(\sqrt{a}+\sqrt{b}\big)^2.$$ □